

Architectural Justification

Layered Architecture — Digital Campus Library Reservation Gateway

Architecture Selection

The gateway is divided into presentation, application, domain and infrastructure layers. The Campus Portal UI handles Student Member and Head Librarian interactions; the Application Controller authenticates and coordinates requests; the Domain Service owns catalogue, circulation, FIFO reservation and fine rules; and infrastructure adapters isolate notification delivery and storage.

Reason 1 — Reliable FIFO reservations

Placing a hold, returning a copy, promoting the queue head and assigning a collection deadline affect shared state. A single domain layer can complete these operations transactionally, preventing concurrent requests from receiving the same position, bypassing an earlier member or offering a copy to the wrong person.

Reason 2 — Appropriate for the system's scale

The system serves one campus catalogue of approximately 50,000 titles. In-process layers are easier to deploy, test and maintain than separate microservices. Client–Server can still describe external access, but it does not organize the server's internal responsibilities.

Security Advantage

The Application Controller acts as the trust boundary. Requests use authenticated, encrypted sessions, and role checks separate Student Member operations from Head Librarian catalogue changes. The UI cannot directly modify holds, loans, fines or ISBN records.

Performance Benefit

Indexed ISBN, title and author fields with pagination avoid scanning or returning the full catalogue. Internal components communicate without network hops, helping searches remain below the 150 ms target. Notification delivery is separated from the return transaction so a slow messaging channel does not delay circulation processing.

Conclusion

Layered Architecture provides the clearest balance of fairness, speed, security and maintainability for the requirements prepared in Lab 1 and planned in Lab 2.